



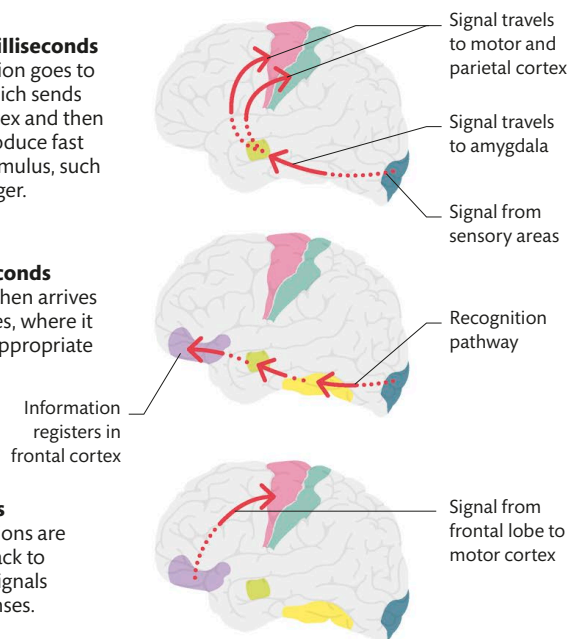
Emotional reactions

Emotional responses evolve over time, from initial protective responses to more considered responses. Imagine a friend leaping out on you: first, you feel shock or fear, but as the brain processes what is happening, you transition to calm. The first stage involves attention being grabbed and the amygdala responding early to prime the conscious brain to “expect” an important perception.

Less than 100 milliseconds
Sensory information goes to the amygdala, which sends signals to the parietal cortex and then to the motor cortex to produce fast reactions to emotional stimulus, such as when fleeing from danger.

100–200 milliseconds
The information then arrives in the frontal lobes, where it becomes conscious and appropriate action is planned.

350 milliseconds
Considered reactions are then conveyed back to the motor cortex, which signals appropriate bodily responses.

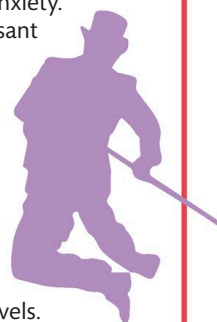


KEY

- Amygdala
- Primary visual cortex
- Frontal cortex
- Fusiform gyrus (face recognition area)
- Motor cortex
- Parietal cortex

SEROTONIN

Alongside dopamine and noradrenaline, serotonin is a neurotransmitter that plays a key role in regulating mood. Although it is not as simple as high serotonin equals happiness and low equals sad, decreases in this molecule are associated with depression and anxiety. Many antidepressant medications act by increasing brain levels of serotonin. Exercise can help, too—for example, taking a brisk walk or dancing can raise serotonin levels.



**EMOTIONS ARE
CONTAGIOUS—
HUMANS MIMIC
EACH OTHER'S
EXPRESSIONS**



Emotions versus moods

Emotions are usually transient—arising from thoughts, activities, or events that act as cues for adaptive behaviors. Moods last for hours, days, or even months. For example, emotion might be experiencing a sudden rush of joy at seeing a friend waiting to greet you compared to a lingering mood of sadness or worry after losing a job. Emotions tend to be expressed in the moment, while moods are not.

ADAPTIVE BEHAVIORS		
EMOTION	POSSIBLE STIMULUS	ADAPTIVE BEHAVIOR
Anger	Challenging behavior from another person	“Fight” reaction prompts dominant and threatening stance or action
Fear	Threat from stronger or dominant person	“Flight” to avoid threat; or act to socially appease the threatening person
Sadness	Loss of loved one	Backward-looking state of mind and passivity, to avoid additional challenge
Disgust	Unwholesome object (e.g., rotting food or unclean surroundings)	Aversion behavior—remove oneself from the unhealthy environment
Surprise	Novel or unexpected event	Attention on object of surprise maximizes sensory input that guides reaction